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EXTENDED ABSTRACT

Evaluating Finetuned Foundation Models for Extreme-Event Representation: Insights from Atmospheric Rivers

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1. Introduction and objectives

Recent studies demonstrate the skill of data-driven weather forecasting, including state-of-the-art AI-based systems. In this context, large weather foundation models (FMs) have emerged as a powerful paradigm. Yet, the ability of such models to generalize across datasets and accurately reproduce high-impact weather extremes remains largely unexplored. We use the Prithvi WxC foundation model, originally trained on MERRA-2 reanalysis data, to assess both its transferability across reanalysis products and its sensitivity to atmospheric rivers (ARs). ARs transport moisture from the tropics to mid-latitudes and drive high-impact events such as extreme precipitation and flooding, making their accurate representation essential. We investigate whether Prithvi WxC can be successfully finetuned on ERA5 to capture the spatiotemporal structure of ARs, evaluating lightweight and diffusion-based decoders for reconstructing and forecasting AR-relevant fields with uncertainty quantification, and benchmark them against established AI systems.

2. Data and Methods

We use ERA5 reanalysis data [1] at 3-hourly temporal resolution (2017–2023) as our finetuning and evaluation dataset. The input comprises 71 atmospheric variables: 65 pressure-level fields at 13 levels (1000–50 hPa) and 6 surface variables, regridded to a 121×240 grid. The original Prithvi-WxC pre-training on MERRA-2 used 160 input fields on native model levels; here, we select pressure levels to enable finetuning on ERA5 while maintaining a comparable vertical representation. For AR characterization, we use integrated water vapor (IWV, here TCWV) and integrated vapor transport (IVT). IWV captures total atmospheric moisture relevant to AR impacts [2], while IVT combines moisture and wind to represent transport intensity and direction. Strong AR events are typically associated with $\mathrm{IVT} > 250\,\mathrm{kg\,m^{-1}\,s^{-1}}$ and $\mathrm{TCWV} > 20\text{–}30\,\mathrm{kg\,m^{-2}}$.

Model and finetuning. Prithvi WxC [4] is a transformer-based FM that learns spatiotemporal atmospheric representations. In all experiments, the pretrained encoder is frozen and only the decoder is optimized. We evaluate two decoder architectures: (i) a lightweight deterministic decoder and (ii) a diffusion-based generative decoder, both operating on the same latent representation (Figure 1).

Evaluation. We evaluate performance in two stages: (i) zero-shot single-step reconstruction, and (ii) autoregressive rollout forecasts over extended lead times. In the rollout setting, we benchmark against

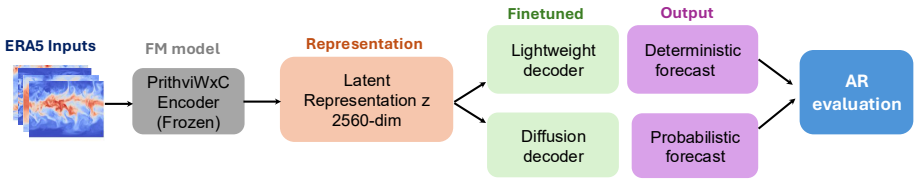

Figure 1. *Finetuning framework. The pretrained encoder is frozen; only the decoder is finetuned on ERA5..*

IFS HRES and operational AI systems including AIFS and AIFS ENS [3]. Outputs are evaluated using AR-focused diagnostics based on IVT and TCWV spatial structure and coherence. For the diffusion decoder, we additionally assess probabilistic forecast characteristics.

3. Preliminary results and expected outcomes

Preliminary experiments demonstrate that the pretrained latent space of Prithvi WxC successfully transfers from MERRA-2 to ERA5. Using a lightweight decoder, the model achieves strong single-step reconstruction skill: IWV (RMSE=3.61 $\mathrm{kg\,m^{-2}}$, ACC=0.85) and IVT (RMSE=8.89 $\mathrm{kg\,m^{-1}\,s^{-1}}$, ACC=0.89), with skill improvements of 80% for TCWV and 65% for IVT over climatological baselines. Spatial skill maps (Figure 2) confirm that the frozen encoder preserves physically meaningful moisture transport patterns. Ongoing work focuses on the diffusion-based decoder for uncertainty quantification and autoregressive rollout benchmarking against operational AI forecasting systems.

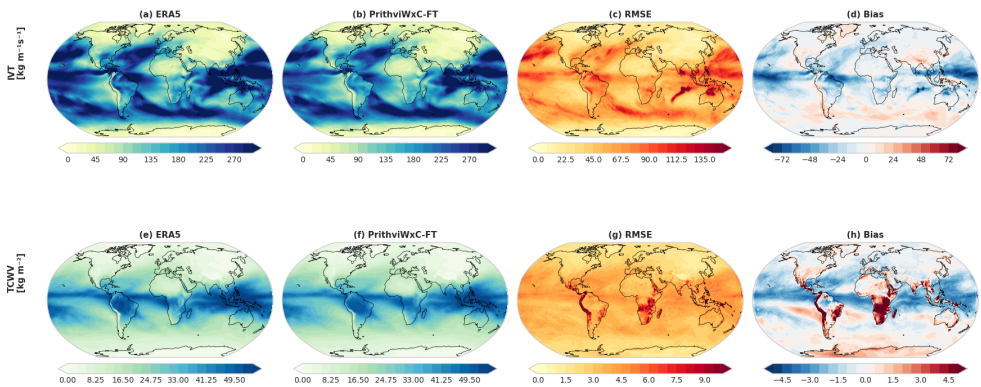

Figure 2. *Spatial prediction skill for IVT and TCWV. Top row: (ab) mean IVT for ERA5 and PrithviWxC-FT, with corresponding (c) RMSE and (d) bias. Bottom row: (ef) mean TCWV for ERA5 and PrithviWxC-FT, with (g) RMSE and (h) bias.*

References

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